

Abstract

College Bus Surveillance using Computer Vision

a) Problem Statement

Ensuring student safety and accurate bus attendance remains a challenge for institutions. Manual roll-calls and paper logs are slow, error-prone, and lack real-time visibility. RFID/card-tap systems can fail when students forget cards and involve additional hardware, making them unreliable and costly. A fully automated, low-cost, real-time monitoring solution is required.

b) Solution Approach

The proposed system offers an intelligent college bus monitoring solution using mobile-based computer vision, GPS tracking, and centralized cloud storage. No specialized hardware such as Raspberry Pi or edge devices is required; instead, modern smartphones serve as the processing unit. The system includes: 1) Centralized student database and web dashboard, 2) Master mobile app for face recognition, 3) Real-time GPS logging and attendance tracking.

c) Technologies Used

Frontend: Flutter. Backend: Python Flask. Firestore stores student data, embeddings, GPS logs, and attendance. Face detection uses Google ML Kit; recognition uses TensorFlow Lite. Additional integrations: camera processing, Google Maps SDK, Directions API, Geolocation API, and OpenStreetMap for route visualization.

d) Base Paper Details

Base Paper 1

“Bus Attendance System Through Facial Recognition by Implementing AI Integration” Authors: Johnson C, Karthick M, Sakthivel P, Sasikala K, Santhi A Published: 2023, *IJRPR*, Vol. 8, Issue 11 Link: <https://www.ijnrd.org/papers/IJNRD2311385.pdf> *IJRPR*

Base Paper 2

“Smart College Bus Tracker: Real-Time Location and ETA Predictor” Authors: M. Sowndharya, Yuvaraj K, Rahul S, Srichandiran S, Sriram S Published: 2025, *IJSRED*, Vol. 8, Issue 3 DOI: IJSRED-V8I3P68 Link: <https://ijsred.com/volume8/issue3/IJSRED-V8I3P68.pdf> *IJSRED*

e) Group No. & Team Members

Group No: 6

- Akash V G – 8
- Fidha Fathima A – 32
- Mohammed Ramshad T S – 44
- Nithya Manoj – 47

f) Time Schedule

Week	Work to be Completed
Week 1	Requirement analysis, scope finalization, architecture diagram, module identification, UI wireframing.
Week 2	DFD/ER diagrams, Flask backend setup, Firestore structure, authentication, REST API development.
Week 3	Web dashboard: admin login, student management, bus assignment, attendance logs, map setup.
Week 4	Flutter app: camera preview, ML Kit detection, UI (login, camera, bus selection).
Week 5	TensorFlow Lite recognition: embedding extraction, comparison, threshold tuning, accuracy tests.
Week 6	Attendance workflow: GPS + timestamp logs, duplicate prevention, offline sync.
Week 7	Real-time GPS tracking, Firestore updates, route plotting, Google Maps/OSM integration.
Week 8	System integration and testing, low-light evaluation, network failure handling, optimization.
Week 9	Security, UI polishing, performance tuning, admin filters, export features.
Week 10	LaTeX report preparation.